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STM32 NUCLEO

TIMER

Elodie PAULY

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Documentations techniques utiles

PM0056 Programming manual
STM32F10xxx/20xxx/21xxx/L1xxxxCortex-M3 programming manual

RM0038 Reference manual
STM32L100xx, STM32L151xx, STM32L152xx and STM32L162xx advanced ARM-based 32-bit MCUs

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Qu'est-ce qu'un timer?

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Un timer est un compteur interne au µC qui peut être incrémenté par un signal d'horloge externe ou interne.

Quel est l'intérêt d'un timer?

La gestion, la mesure et la maîtrise du temps!

En effet, dans beaucoup d'applications, on a besoin de mesurer des durées, de générer des signaux bien définis temporellement etc …

Sans timer, cela serait bien difficile, car dépendant du temps d'exécution des instructions. Cela nécessiterait un contrôle délicat des boucles.

Timers sur le STM32

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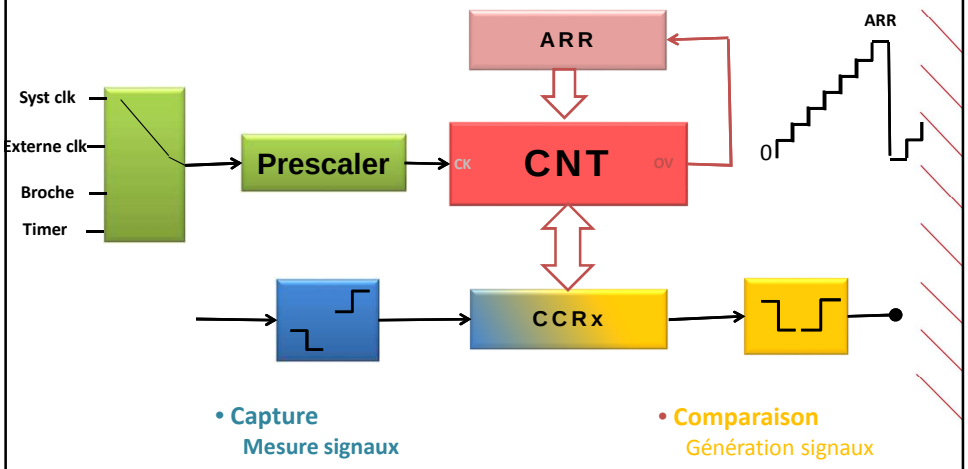
Table 6. Timer feature comparison

Timer	Counter resolution	Counter type	Prescaler factor	DMA request generation	Capture/compare channels	Complementary outputs
TIM2, TIM3, TIM4	16-bit	Up, down, up/down	Any integer between 1 and 65536	Yes	4	No
TIM5	32-bit	Up, down, up/down	Any integer between 1 and 65536	Yes	4	No
TIM9	16-bit	Up, down, up/down	Any integer between 1 and 65536	No	2	No
TIM10, TIM11	16-bit	Up	Any integer between 1 and 65536	No	1	No
TIM6, TIM7	16-bit	Up	Any integer between 1 and 65536	Yes	0	No

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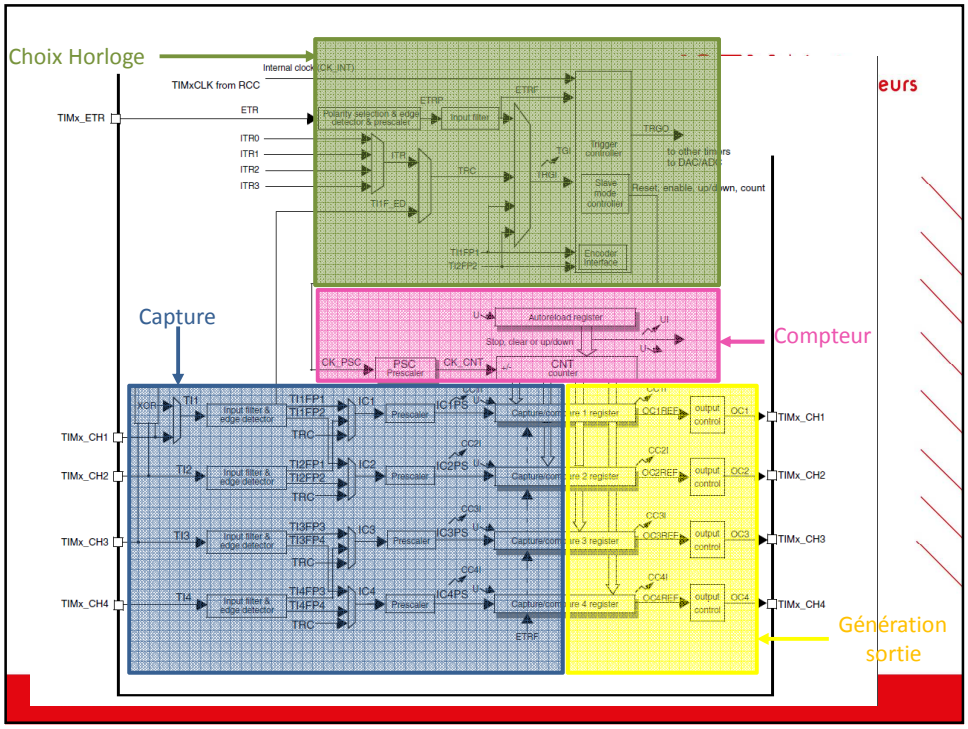
Structure simplifiée du timer STM32

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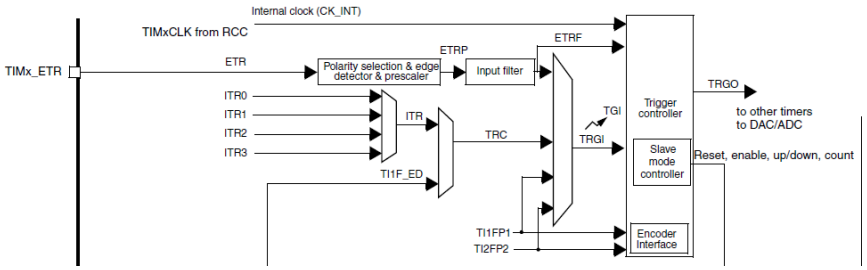
- **Capture**
  Mesure signaux
- **Comparaison**
  Génération signaux

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Choix de l'horloge

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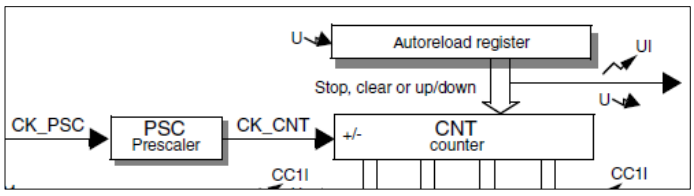


Nous avons le choix:

- Horloge interne CK_INT
- Horloge externe mode 2 (TIMx_ETR)
- Trigger interne (ITR)
- Horloge externe mode 1 (Tix)

Compteur

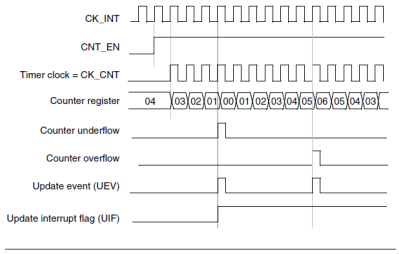
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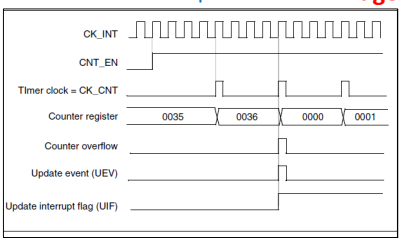
Mode de fonctionnement du compteur

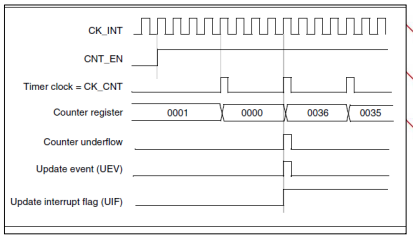
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Center-aligned mode (up/down counting)



Mode Up counter **Edge-aligned mode** Mode down counter





Configuration du Mode Base Time Unit

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1- Activer le périphérique TIMx sur le périphérique RCC

2- Configurer l'horloge du compteur
- configurer le prescaler

3- Configurer le mode du compteur:
- Edge Aligned Mode Down
- Edge Aligned Mode Up
- Center Aligned Mode

4- Donner la valeur de départ (down counting) ou de rechargement (up counting)

5- Configurer le cas échéant les interruptions

6- Lancer le timer

17.4.1 TIMx control register 1 (TIMx_CR1)

Address offset: 0x00

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved						CKD[1:0]		ARPE	CMS		DIR	OPM	URS	UDIS	CEN
						rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bit 7 **ARPE**: Auto-reload preload enable
0: TIMx_ARR register is not buffered
1: TIMx_ARR register is buffered

Bits 6:5 **CMS**: Center-aligned mode selection
00: Edge-aligned mode. The counter counts up or down depending on the direction bit (DIR).
01: Center-aligned mode 1. The counter counts up and down alternatively. Output compare interrupt flags of channels configured in output (CCxS=00 in TIMx_CCMRx register) are set only when the counter is counting down.
10: Center-aligned mode 2. The counter counts up and down alternatively. Output compare interrupt flags of channels configured in output (CCxS=00 in TIMx_CCMRx register) are set only when the counter is counting up.
11: Center-aligned mode 3. The counter counts up and down alternatively. Output compare interrupt flags of channels configured in output (CCxS=00 in TIMx_CCMRx register) are set both when the counter is counting up or down.
Note: It is not allowed to switch from edge-aligned mode to center-aligned mode as long as the counter is enabled (CEN=1)

Bit 4 **DIR**: Direction
0: Counter used as upcounter
1: Counter used as downcounter
Note: This bit is read only when the timer is configured in Center-aligned mode or Encoder mode

Bit 0 **CEN**: Counter enable
0: Counter disabled
1: Counter enabled
Note: External clock, gated mode and encoder mode can work only if the CEN bit has been previously set by software. However trigger mode can set the CEN bit automatically by hardware.
CEN is cleared automatically in one-pulse mode, when an update event occurs.

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Registre de Contrôle TIMx_CR1

Changement de valeur Auto Reload

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Figure 91. Counter timing diagram, Update event when ARPE=0 (TIMx_ARR not preloaded)

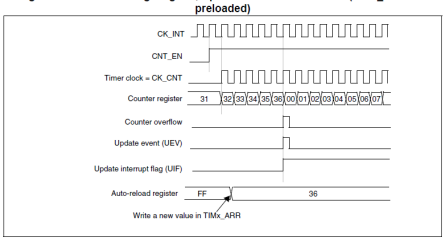
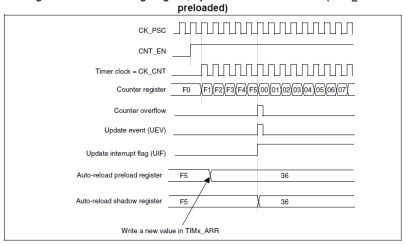


Figure 92. Counter timing diagram, Update event when ARPE=1 (TIMx_ARR preloaded)



Lorsque la valeur de TIMx_ARR n'est pas pré-chargée (ARPE = 0) le changement de valeur de TIMx_ARR n'est prise en compte qu'au prochain évènement.

17.4.10 TIMx counter (TIMx_CNT)

Address offset: 0x24

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CNT[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 15:0 **CNT[15:0]**: Counter value

17.4.11 TIMx prescaler (TIMx_PSC)

Address offset: 0x28

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
PSC[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 15:0 **PSC[15:0]**: Prescaler value

The counter clock frequency CK_CNT is equal to f_{CK_PSC} / (PSC[15:0] + 1).

PSC contains the value to be loaded in the active prescaler register at each update event.

$$F_{CK_CNT} = \frac{F_{CK_PSC}}{PSC + 1}$$

17.4.12 TIMx auto-reload register (TIMx_ARR)

Address offset: 0x2C

Reset value: 0xFFFF

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
ARR[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 15:0 **ARR[15:0]**: Auto-reload value

ARR is the value to be loaded in the actual auto-reload register.

Refer to the *Section 17.3.1: Time-base unit on page 380* for more details about ARR update and behavior.

The counter is blocked while the auto-reload value is null.

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Indicateurs d'évènements et autorisation d'interruptions

17.4.5 TIMx status register (TIMx_SR)

Address offset: 0x10

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Reserved			CC4OF	CC3OF	CC2OF	CC1OF	Reserved		TIF	Res	CC4IF	CC3IF	CC2IF	CC1IF	UIF
			rc_w0	rc_w0	rc_w0	rc_w0			rc_w0		rc_w0	rc_w0	rc_w0	rc_w0	rc_w0

17.4.4 TIMx DMA/Interrupt enable register (TIMx_DIER)

Address offset: 0x0C

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Res.	TDE	Res	CC4DE	CC3DE	CC2DE	CC1DE	UDE	Res.	TIE	Res	CC4IE	CC3IE	CC2IE	CC1IE	UIE
	rw		rw	rw	rw	rw	rw		rw		rw	rw	rw	rw	rw

Bit 0 **UIE**: Update interrupt enable

0: Update interrupt disabled

1: Update interrupt enabled

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Interruption timer

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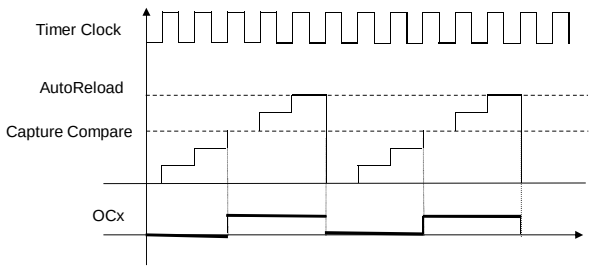
1. Configuration du Périphérique Timer
   2.1 Autoriser la génération d'interruption ou d'évènement
   2.2 Remettre à 0 le bit d'état de l'interruption

2. Configuration du NVIC
   1.1 Déterminer le N° de l'interruption
   1.2 Activer l'interruption
   1.3 Gérer les priorités si besoin

1. Ne pas oublier d'écrire et de déclarer la routine d'interruption dans ***stm32l1xx_it.h*** et ***stm32l1xx_it.c***
   3.1 Reprendre le nom identique à la table de vecteur
   3.2 Dans la routine d'interruption, ne pas oublier de remettre le bit d'état de l'interruption à 0 afin de permettre la détection d'un nouvel évènement

Mode PWM

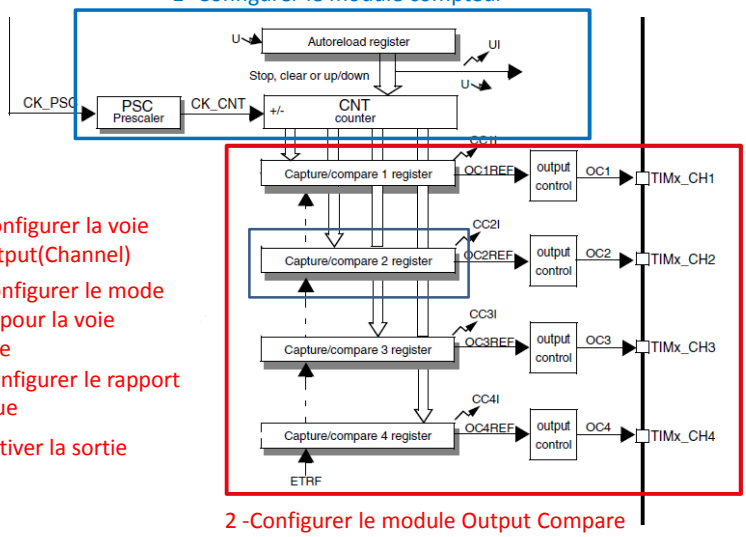
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Un signal PWM (Pulse Width Modulation) est un signal périodique dont le rapport cyclique (Durée à l'état haut/Période) est variable.

Mode PWM

1 -Configurer le module compteur



2.1 Configurer la voie en output(Channel)

2.2 Configurer le mode PWM pour la voie désirée

2.3 Configurer le rapport cyclique

2.4 Activer la sortie

2 -Configurer le module Output Compare

Mode PWM

1- Configurer le module Timer
- 1.1 Activer le périphérique TIMx
- 1.2 Configurer l'horloge
- 1.3 Configurer le prescaler
- 1.4 Configurer le mode du compteur
- 1.5 Configurer la période du signal PWM désiré

2- Configurer le module Input/Output Compare
- 2.1 Configurer le module en Ouptut compare c'est-à-dire mettre la voie choisie en mode Ouptut
- 2.2 Mettre en mode PWM (choix entre 2 modes)
- 2.3 Configurer le rapport cyclique
- 2.4 Activer la sortie PWM qu'on veut utiliser

(Ne pas oublier aussi de configurer la broche en mode Alternative Function Push Pull dans le GPIO correspondant)

3- Démarrer le timer : le signal PWM sera automatiquement généré par le compteur sans aucune action de votre part

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ATTENTION A NE PAS OUBLIER DE:
- CONFIGURER LE RCC pour activer les périphériques mis en jeux
- ET LE GPIO pour configurer les pins utilisées par les autres périphériques que le GPIO

Mode PWM: Registres de configuration

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17.4.7 TIMx capture/compare mode register 1 (TIMx_CCMR1)

Address offset: 0x18

Reset value: 0x0000

The channels can be used in input (capture mode) or in output (compare mode). The direction of a channel is defined by configuring the corresponding CCxS bits. All the other bits of this register have a different function in input and in output mode. For a given bit, OCxx describes its function when the channel is configured in output, ICxx describes its function when the channel is configured in input. So you must take care that the same bit can have a different meaning for the input stage and for the output stage.

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
OC2CE	OC2M[2:0]			OC2PE	OC2FE	CC2S[1:0]		OC1CE	OC1M[2:0]			OC1PE	OC1FE	CC1S[1:0]	
IC2F[3:0]				IC2PSC[1:0]				IC1F[3:0]				IC1PSC[1:0]			
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Pour la Voie 2 (Channel 2)

Pour la Voie 1 (Channel 1)

Mode PWM: Registres de configuration

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Bits 6:4 **OC1M**: Output compare 1 mode

These bits define the behavior of the output reference signal OC1REF from which OC1 and OC1N are derived. OC1REF is active high whereas OC1 and OC1N active level depends on CC1P and CC1NP bits.
000: Frozen - The comparison between the output compare register TIMx_CCR1 and the counter TIMx_CNT has no effect on the outputs.(this mode is used to generate a timing base).
001: Set channel 1 to active level on match. OC1REF signal is forced high when the counter TIMx_CNT matches the capture/compare register 1 (TIMx_CCR1).
010: Set channel 1 to inactive level on match. OC1REF signal is forced low when the counter TIMx_CNT matches the capture/compare register 1 (TIMx_CCR1).
011: Toggle - OC1REF toggles when TIMx_CNT=TIMx_CCR1.
100: Force inactive level - OC1REF is forced low.
101: Force active level - OC1REF is forced high.
110: PWM mode 1 - In upcounting, channel 1 is active as long as TIMx_CNT<TIMx_CCR1 else inactive. In downcounting, channel 1 is inactive (OC1REF='0) as long as TIMx_CNT>TIMx_CCR1 else active (OC1REF=1).
111: PWM mode 2 - In upcounting, channel 1 is inactive as long as TIMx_CNT<TIMx_CCR1 else active. In downcounting, channel 1 is active as long as TIMx_CNT>TIMx_CCR1 else inactive.

Note: In PWM mode 1 or 2, the OCREF level changes only when the result of the comparison changes or when the output compare mode switches from "frozen" mode to "PWM" mode.

Mode PWM: Registres de configuration

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17.4.13 TIMx capture/compare register 1 (TIMx_CCR1)

Address offset: 0x34

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CCR1[15:0]															
rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw	rw

Bits 15:0 **CCR1[15:0]**: Capture/Compare 1 value

If channel CC1 is configured as output:
CCR1 is the value to be loaded in the actual capture/compare 1 register (preload value).
It is loaded permanently if the preload feature is not selected in the TIMx_CCMR1 register (bit OC1PE). Else the preload value is copied in the active capture/compare 1 register when an update event occurs.
The active capture/compare register contains the value to be compared to the counter TIMx_CNT and signaled on OC1 output.

If channel CC1is configured as input:
CCR1 is the counter value transferred by the last input capture 1 event (IC1).

FIXE LE RAPPORT CYCLIQUE DU SIGNAL PWM GENERE

Mode PWM: Registres de configuration

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17.4.9 TIMx capture/compare enable register (TIMx_CCER)

Address offset: 0x20

Reset value: 0x0000

15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
CC4NP	Res.	CC4P	CC4E	CC3NP	Res.	CC3P	CC3E	CC2NP	Res.	CC2P	CC2E	CC1NP	Res.	CC1P	CC1E
rw		rw	rw	rw		rw	rw	rw		rw	rw	rw		rw	rw

Bit 0 **CC1E**: *Capture/Compare 1 output enable.*

CC1 channel configured as output:
0: Off - OC1 is not active
1: On - OC1 signal is output on the corresponding output pin

CC1 channel configured as input:
This bit determines if a capture of the counter value can actually be done into the input capture/compare register 1 (TIMx_CCR1) or not.
0: Capture disabled
1: Capture enabled

TIMERS sur µC

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Ceci est un **cours non exhaustif** sur les usages et les fonctionnalités du timer sur le microcontrôleur et en particulier sur le STM32L152RE

Il vous appartient **d'aller lire la documentation technique** du composant et en particulier du timer afin d'améliorer et de configurer le timer à votre convenance.

Même si ce cours est appliqué au STM32L152RE, l'usage et le fonctionnement d'un timer reste sensiblement le même sur n'importe quel autre microcontrôleur. Il vous faudra en revanche trouver les registres équivalents.

Documentation Technique de référence: **RM0038 Reference manual**
STM32L100xx, STM32L151xx, STM32L152xx and STM32L162xx
advanced ARM-based 32-bit MCUs
TIMER: Pages 379 à 438